

Junhao Huang

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Postal Address: Singapore Management University, 81 Victoria Street, Singapore 188065



Work

- **Singapore Management University**

Research Scientist

Supervisor: Dr. Haiyang Xue

Jan. 2026-Now

- **City University of Hong Kong**

Postdoc

Supervisor: Prof. Ray C.C. Cheung

Oct. 2025-Dec. 2025

Education

- **Beijing Normal-Hong Kong Baptist University**

PhD Degree of Computer Science (Hong Kong Baptist University)

Supervisor: Dr. Donglong Chen

Sep. 2021-Jul. 2025

- **Nanjing University of Aeronautics and Astronautics**

Master Degree of Cyberspace Security

Supervisor: Prof. Zhe Liu

Sep. 2018-Jun. 2021

- **Nanjing University of Aeronautics and Astronautics**

Bachelor Degree of Computer Science and Technology

GPA: 3.7

Sep. 2014-Jun. 2018

Research/Visiting Activities

- **Visiting Scholar, Electrical Engineering**

City University of Hong Kong (Prof. Ray C. C. Cheung)

Hong Kong, China

Jul. 2023-Dec. 2023

- **Visiting Scholar, Cyberspace Security**

Wuhan University (Prof. Debiao He)

Wuhan, China

Sep. 2019-Jan. 2020

Scientific Services

I have been actively participating as a Program Committee/reviewer for the following conferences/journals:

- *CCF International Conference on Service Science (CCF ICSS)*
- *ACM Transactions on Architecture and Code Optimization (TACO).*
- *Journal of Cryptographic Engineering (JCEN).*
- *The Journal of Supercomputing.*

I have also been actively participating as a Artifact Evaluation Committee in the following IACR conferences:

- **IACR CHES 2026 Artifact Evaluation Committee** Antalya, Türkiye
International Association for Cryptologic Research (IACR) Oct. 2025-Oct. 2026
- **IACR CHES 2025 Artifact Evaluation Committee** Kuala Lumpur, Malaysia
International Association for Cryptologic Research (IACR) Oct. 2024-Oct. 2025
- **IACR CHES 2024 Artifact Evaluation Committee** Halifax, Canada
International Association for Cryptologic Research (IACR) Oct. 2023-Oct. 2024

Research Interests

- Modular Arithmetic, Lattice-based Cryptography, Fully Homomorphic Encryption, ARM & RISC.

Representative Publications (Total Publications: 15)

1. **Junhao Huang**, Haosong Zhao, Jipeng Zhang, Wangchen Dai, Lu Zhou, Çetin Kaya Koç, Ray C.C. Cheung, Donglong Chen*, Yet another Improvement of Plantard Arithmetic for Faster Kyber on Low-end 32-bit IoT Devices, *IEEE Transactions on Information Forensics & Security*, 2024.
2. **Junhao Huang**, Alexandre Adomnicăi, Jipeng Zhang, Wangchen Dai, Yao Liu, Ray C. C. Cheung, Çetin Kaya Koç, Donglong Chen*, Revisiting Keccak and Dilithium Implementations on ARMv7-M, *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2024.
3. **Junhao Huang**, Jipeng Zhang, Haosong Zhao, Zhe Liu, Ray C. C. Cheung, Çetin Kaya Koç, Donglong Chen*, Improved Plantard Arithmetic for Lattice-based Cryptography, *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2022.
4. **Junhao Huang**, Zhe Liu*, Zhi Hu, and Johann Großschädl, Parallel Implementation of SM2 Elliptic Curve on Intel Processor with AVX2, *Australasian Conference on Information Security and Privacy - ACISP 2020*.
5. Zewen Ye, **Junhao Huang**, Tianshun Huang, Yudan Bai, Jinze Li, Hao Zhang, Guangyan Li, Donglong Chen, Ray CC Cheung, Kejie Huang, PQNTRU: Acceleration of NTRU-based Schemes via Customized Post-Quantum Processor[J]. *IEEE Transactions on Computers*, 2025
6. Haosong Zhao, **Junhao Huang**, Zihang Chen, Kunxiong Zhu, Donglong Chen, Zhuoran Ji, Hongyuan Liu, VESTA: A Secure and Efficient FHE-based Three-Party Vectorized Evaluation System for Tree Aggregation Models[C]. *ACM SIGMETRICS*, 2025
7. Jipeng Zhang, Yuxing Yan, **Junhao Huang**, Çetin Kaya Koç*, Optimized Software Implementation of Keccak, Kyber, and Dilithium on $RV\{32,64\}IM\{B\}\{V\}$, *IACR Transactions on Cryptographic Hardware and Embedded Systems*, 2025.
8. Jipeng Zhang, **Junhao Huang**, Lirui Zhao, Donglong Chen, Çetin Kaya Koç*, ENG25519: Faster TLS 1.3 handshake using optimized X25519 and Ed25519, *Usenix Security*, 2024.
9. Jipeng Zhang, **Junhao Huang**, Zhe Liu*, Sujoy Sinha Roy, Time-memory Trade-offs for Saber on Memory-constrained RISC-V, *IEEE Transactions on Computers*, 2022.

10. Xinyi Ji, Jiankuo Dong, **Junhao Huang**, Zhijian Yuan, Wangchen Dai, Fu Xiao, Jingqiang Lin, ECO-CRYSTALS: Efficient Cryptography CRYSTALS on Standard RISC-V ISA, *IEEE Transactions on Computers*, 2024.

Academic Referee

1. **Dr. Donglong Chen**: Associate Professor of Beijing Normal-Hong Kong Baptist University, donglongchen@uic.edu.cn
2. **Prof. Ray C.C. Cheung**: Associate Provost, Professor of City University of Hong Kong, r.cheung@cityu.edu.hk
3. **Prof. Çetin Kaya Koç**: IEEE Life Fellow, Co-founder of CHES, University of California Santa Barbara, cetinkoc@ucsb.edu